Bangladesh AI Policy & Innovation Lab (BAPIL)

Blueprint Summary

Summary of the BAPIL Technical Blueprint (Version 1.0)

Version 1.0

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Repository: https://github.com/md-naim-hassan-saykat/bapil

This summary presents the main design choices, safeguards, delivery plan, strategic roadmap to 2041 and decisions set out in the BAPIL Technical Blueprint (five parts, 21 chapters, 22 annexes). Figure and table numbers follow the full Blueprint. The Blueprint and this summary are independent research documents and do not represent any official position. BAPIL is a proposal: no institution has been created and no funding has been approved.

1. What BAPIL Is, and Why Now

The Bangladesh AI Policy & Innovation Lab (BAPIL) is a proposed independent research and policy-intelligence platform that helps Bangladesh use artificial intelligence for evidence-based policymaking, responsible AI governance, Bangla-language technology and public-sector innovation. The White Paper (Version 1.0) set out the vision; the Technical Blueprint turns it into a buildable plan.

The timing is favourable. As reported, Bangladesh converted its 2025 data ordinances into the Personal Data Protection Act, 2026 and the National Data Management Act, 2026 in April 2026 (which provisions are in force is still to be confirmed), revised its cyber-security law, released a draft National AI Policy 2026-2030 for consultation in January 2026, and announced that drafting of a national AI policy would begin in the fourth quarter of 2026 [85, 86, 87, 88, 91]. A concrete, open blueprint with working code can be a technical input to that process.

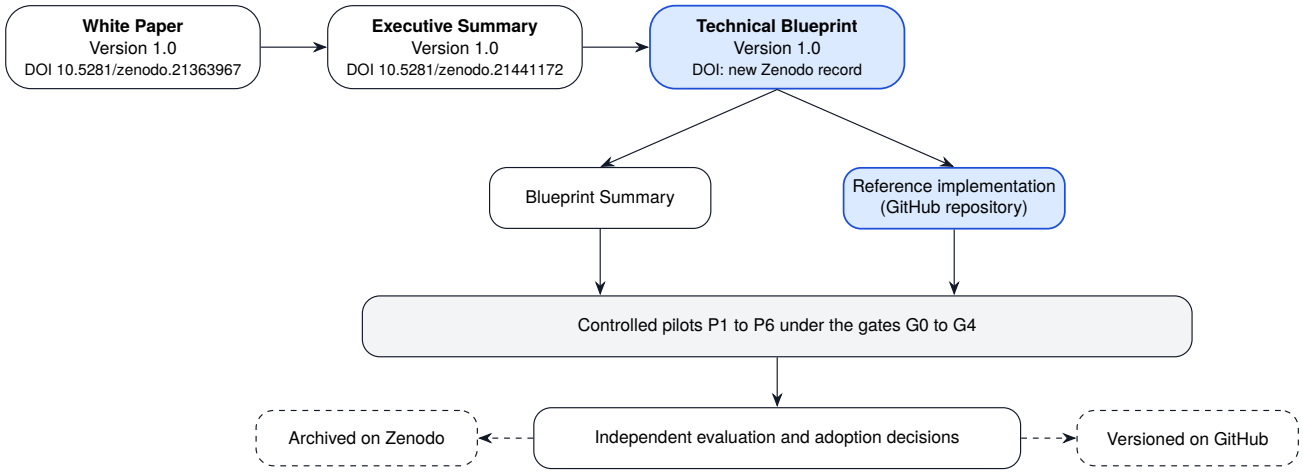


Figure 1. The BAPIL document family: from the White Paper to a tested implementation and controlled pilots.

2. What the Blueprint Specifies

1. **26 requirements** (functional, non-functional and governance), each traced to components, code and tests.
2. **An institution:** a university-anchored lab with a government MoU, a multi-stakeholder Steering Board, an independent Ethics and Risk Review Committee (ERRC) with a binding veto, and a Technical Architecture Board.
3. **A reference architecture** of twelve open, replaceable components.
4. **Data rules:** four data classes, a public-data-first policy and an inventory of official sources.
5. **A trustworthy AI design:** bilingual search and answers in which every sentence cites an official source, with abstention when evidence is weak.
6. **An open benchmark,** BanglaPolicyQA, and a staged, evidence-gated path to a Bangla Policy Foundation Model.
7. **Six pilots** with users, data, risk tier, scope and exit criteria.
8. **Security and hosting** in Bangladesh, with controls for AI-specific threats.
9. **An assurance system:** risk tiers T0 to T3, lifecycle gates G0 to G5, model cards and a public transparency register.
10. **A small, tested reference implementation** (a research prototype, not a production system) issued on GitHub with the Blueprint, a 36-month delivery plan, staffing and training, financing and procurement clauses, a logical framework, KPIs, a risk register and a research agenda.
11. **A strategic roadmap for 2026 to 2041 and beyond,** with five horizons, indicative targets, scenarios and fixed review points.
12. **Six decisions** for the authorities.

13. **Evidence and readiness annexes:** the evidence status of every key claim, budget reconciliation and scenarios, an evaluation protocol with sample sizes, the status and limits of each governance body, the preconditions for starting, risk monitoring, and a register of unresolved items.

3. Institution and Governance

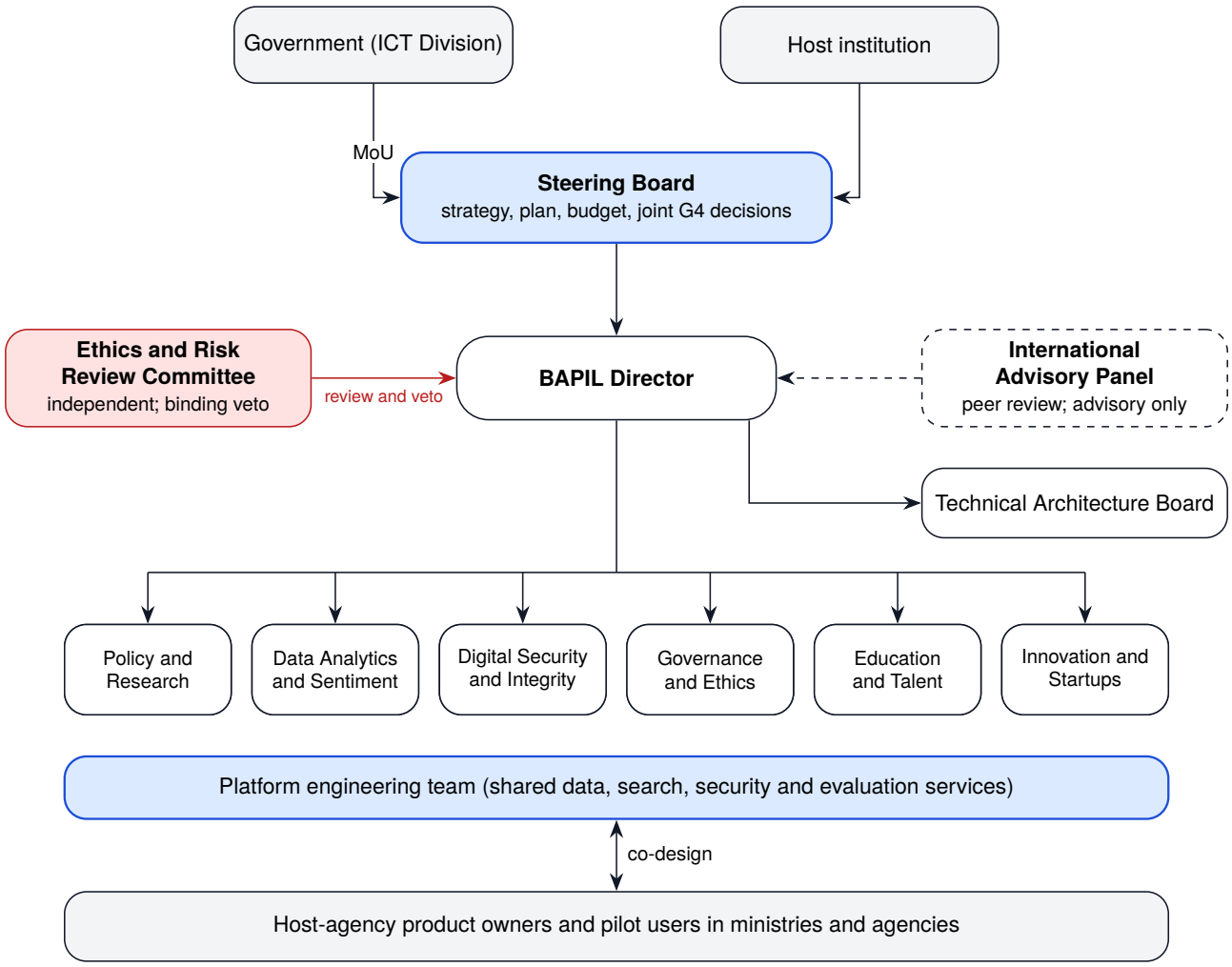


Figure 3. Proposed institutional and governance model (Option C: university-anchored lab with a government MoU).

A hybrid model (Option C in the Blueprint) combines academic independence with a direct channel to government. Successful tools are transferred to the responsible agency through a documented adoption decision; BAPIL never operates production government systems.

4. Architecture and AI Design

The core service is a **bilingual, citation-grounded policy search and answer system** over officially published documents: laws, gazettes, budgets, plans and statistics. Text is normalised for Bangla, queries are expanded across Bangla and English, passages are retrieved and checked, and an answer is given only when evidence is sufficient. Every query, source and output is written to a tamper-evident audit log. Models are open-weight and, for any non-public data, hosted in Bangladesh.

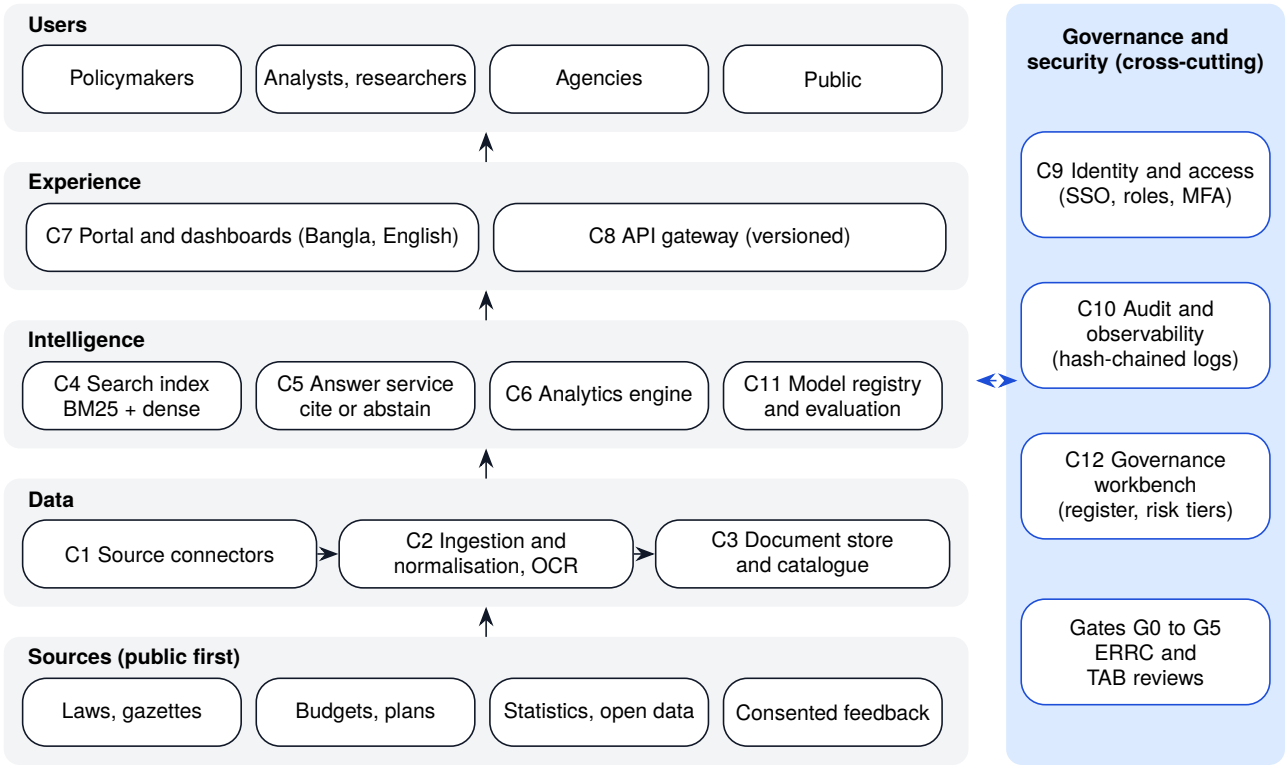


Figure 4. BAPIL reference architecture: twelve components in four layers, with cross-cutting governance and security.

5. Pilot Portfolio

Pilot	Data	Tier	First-year result
P1 Bangladesh Policy Search Engine	Public documents	T1 or T2	Pilot with 30 or more users in two to three agencies
P2 Budget Intelligence Dashboard	Published budget documents	T1	Five fiscal years of allocations as open data
P3 Citizen Feedback Analyser	Consented, de-identified submissions	T2	Aggregate summaries of one consultation after a DPIA
P4 Bangla Policy Foundation Model, Stages 0 and 1	Governed public corpus	T1	BanglaPolicyQA benchmark v0 and baselines
P5 National AI Readiness Observatory	Public indicators	T1	Indicator framework and first report
P6 Information-integrity research	Public research datasets	T2	Methods review; no monitoring of individuals

6. Safeguards

- **Public data first;** personal data only with consent, a DPIA and ERRC approval; sensitive personal data never.
- **Advice, not decisions;** a person reviews every output in advisory use, and no system decides about individuals.
- **No surveillance;** social scoring, mass biometric surveillance and profiling of political or religious opinions

are prohibited by charter.

- **Risk tiers and gates** decide how much control a system needs and when it may move forward.
- **Openness:** code, benchmarks, evaluation reports and the transparency register are published.

Table 39. Risk tiers and the minimum controls each tier requires

Tier	Typical systems	Minimum required controls	Who approves
T0 Prohibited	Social scoring; mass or indiscriminate biometric surveillance; profiling people's political or religious opinions; deceptive synthetic media of real people; manipulation exploiting vulnerabilities	Not built, not piloted, not supported by BAPIL under any arrangement	Excluded by charter
T1 Low	Research tools and search over public documents; dashboards of published data	AI system record; model or data card; evaluation report; audit logging; public listing in the transparency register	Unit lead and TAB
T2 Limited	Advisory tools used by officials; public-facing information tools; aggregate analysis of de-identified feedback	All T1 controls plus: ERRC review; red-team test; human review of every output in use; user training; incident procedure; quarterly monitoring report	ERRC and Steering Board
T3 High	Any system that informs decisions about specific people or entities, processes personal data at scale, or is used in enforcement	BAPIL does not build T3 operational systems. Research only, with: DPIA, legal review, independent audit, ERRC approval, and a published impact assessment	ERRC (binding) and host-agency legal review

7. Delivery Plan (36 Months)

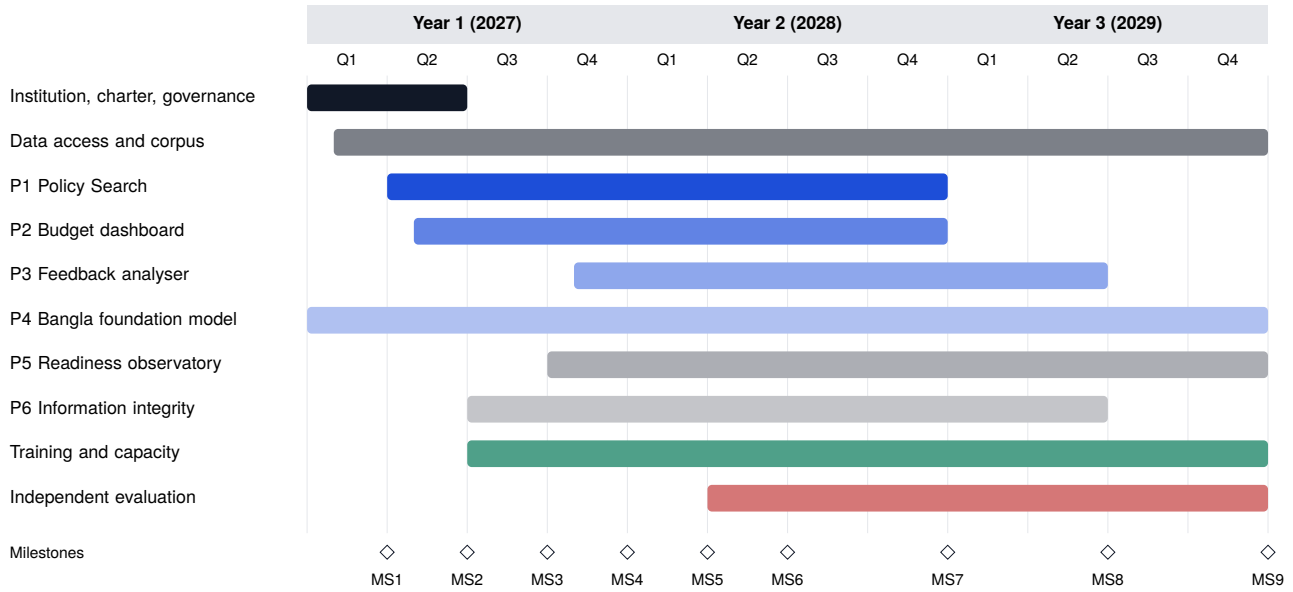


Figure 12. Indicative 36-month delivery roadmap (Month 1 is the month the MoU takes effect; indicative start January 2027).

- **Before month 1:** MoU, host agreement and Year 1 funding in place.
- **Month 3:** Director appointed; Steering Board, ERRC and TAB in place; charter adopted.
- **Month 6:** feasibility and costing study; data access confirmed; benchmark protocol agreed.
- **Month 12:** P1 pilot running with 30 or more users; P2 open budget dataset published; 100 officials trained.
- **Month 24:** adoption decisions on P1 and P2 with host agencies; first AI readiness report; 300 officials trained.
- **Month 36:** final independent evaluation; sustainability plan; Blueprint 2.0.

8. Strategic Roadmap 2026-2041 and Beyond

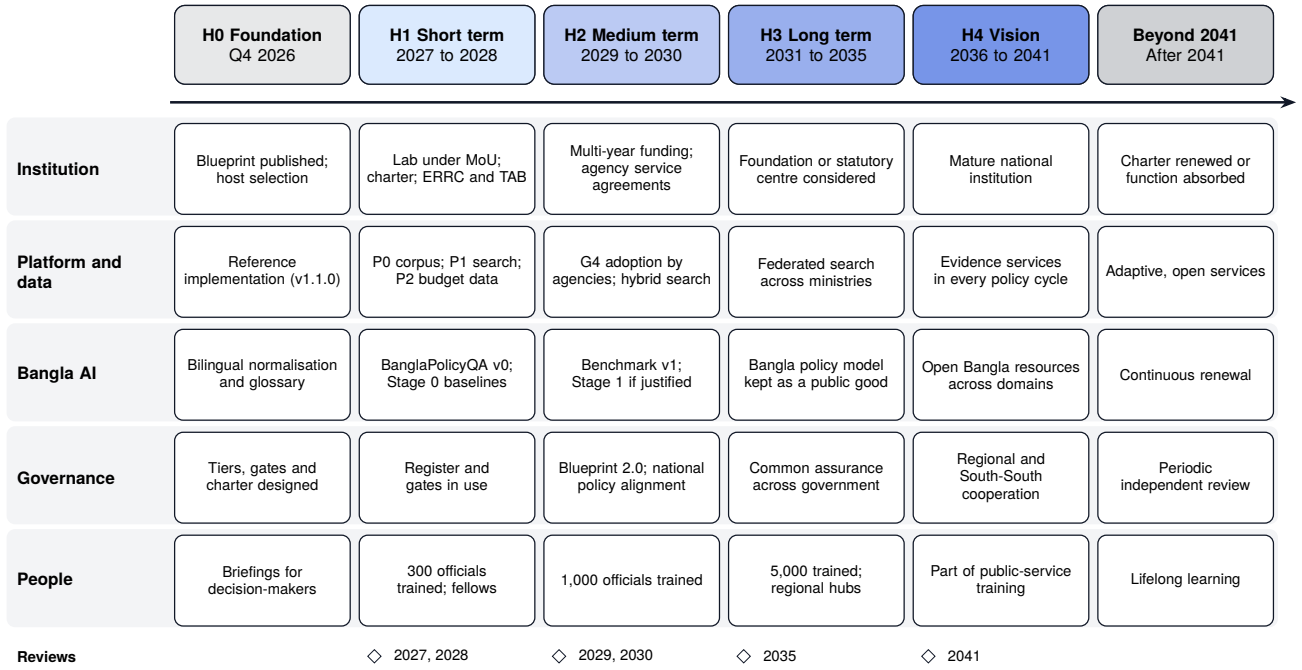


Figure 13. Strategic roadmap, 2026 to 2041 and beyond: horizons, workstreams and review points.

The 36-month plan is the first step of a longer strategy that connects BAPIL's pilots to the national AI policy cycle and to the Perspective Plan 2041 and ICT Masterplan 2041 [29, 30, 88].

- **H0 Foundation (Q4 2026):** publish the Blueprint and code; brief the authorities; offer input to national AI policy drafting; select a host institution.
- **H1 Short term (2027 to 2028):** prove value with public data; P1 and P2 pass their pilot gates; about 300 officials trained; mid-term evaluation.
- **H2 Medium term (2029 to 2030):** agencies adopt successful tools; benchmark v1; Blueprint 2.0; contribution to the post-2030 national AI policy; about 1,000 officials trained.
- **H3 Long term (2031 to 2035):** a national policy-intelligence network on shared standards; common assurance methods across government; regional training hubs; decision on institutional form.
- **H4 Vision (2036 to 2041):** evidence services embedded in policy cycles; open Bangla language resources across domains; regional and South-South cooperation; independent review in 2041.
- **Beyond 2041:** the charter is renewed only if the 2041 review confirms continuing public value; otherwise BAPIL closes or merges, with open archives preserved.

Targets are indicative and revised at fixed reviews (2027, 2028, 2029, 2030, 2035 and 2041) and whenever a major event occurs. In every scenario, the core (governed corpus, cited search, benchmark and governance) is protected, and safeguards never weaken as BAPIL grows.

Table 45. Strategic horizons, 2026 to 2041 and beyond

Horizon	Period	Strategic aim	Priority actions	Institutional form	Signal to move on
H0 Foundation	Q4 2026	Prepare a safe, open start	Publish this Blueprint and code; brief the ICT Division and custodians; offer input to national AI policy drafting; select host institution	Independent research initiative	Decisions D1, D2 and D4 taken

Horizon	Period	Strategic aim	Priority actions	Institutional form	Signal to move on
H1 Short term	2027 to 2028	Prove value with public data	Charter, ERRC and register; P1 and P2 pilots; BanglaPolicyQA v0; 300 officials trained; mid-term evaluation	University-anchored lab under MoU (Option C)	P1 and P2 pass G3; positive mid-term evaluation
H2 Medium term	2029 to 2030	Adopt, scale and align with national policy	G4 adoption of P1 and P2 by agencies; BPFM Stage 1 if justified; benchmark v1; Blueprint 2.0; align with the national AI policy cycle ending 2030	Lab with multi-year core funding; agency service agreements	Tools in operational use; sustained funding agreed
H3 Long term	2031 to 2035	Build a national policy-intelligence network	Federated search across ministries and divisions; Bangla policy model maintained as a public good; standard assurance used across government; regional training hubs	Consider an independent public-interest foundation (Option D) or statutory centre	Assurance methods used beyond BAPIL; network in most ministries
H4 Vision	2036 to 2041	Trusted AI-enabled policy system for a knowledge economy	Evidence services embedded in policy cycles; open Bangla language resources across domains; regional and South-South cooperation; contribution to Perspective Plan 2041 and ICT Masterplan 2041 goals	Mature national institution with international partnerships	Independent review at 2041 confirms public value
Beyond 2041	After 2041	Keep the capability adaptive and accountable	Periodic renewal of mission and charter; sunset or merger if the function is fully absorbed by government; preserve open archives	Determined by the 2041 review	Charter renewal decision

Note. Horizons H1 and H2 correspond to the 36-month roadmap of Chapter 15 (indicative start January 2027). Later horizons are strategic intentions, reviewed at the points in Table 48.

9. People, Resources and Financing

The core team grows from about 9 to 19 full-time equivalents over three years. Pilot computing needs are modest (two to three servers and two GPU inference servers, doubled in Year 2 for staging and recovery), with any model adaptation using shared national compute. An indicative budget built from stated assumptions comes to about Tk 20.4 crore over 36 months (Tk 4.7, 7.5 and 8.2 crore in Years 1 to 3, about USD 1.65 million), including 8% annual escalation, host overheads and a 5% contingency, with people at about half of the total. Low and high scenarios give about Tk 17.9 and 25.3 crore. Some costs are not yet included (engagement events, survey data collection, travel, open-access fees and staff computers); with an allowance for them and a 10% contingency, the estimate is about Tk 22.2 crore. It is a planning estimate, not approved funding; the feasibility and costing study in month 6 confirms or replaces each assumption. Funding is diversified, with a growing share of core public funding and agency service agreements over time; no single non-government funder provides more than 30% of the budget. Model procurement clauses protect data, guarantee the right to evaluate and prevent lock-in.

10. Reference Implementation and Research

The repository contains a tested, dependency-free Python implementation of the core ideas: Bangla normalisation, bilingual BM25 search, cited extractive answers with abstention, risk-tier assessment, an AI system register with transparency export, evaluation metrics and a hash-chained audit log, with a command line and a small JSON API. Continuous-integration workflows are configured to run the tests and rebuild the published documents from source. It has not been deployed or tested with users; the evaluation thresholds in the Blueprint are proposed targets, with a measurement protocol, not results. A research agenda of six questions, from Bangla retrieval to institutional design, leads to open datasets, evaluation reports and peer-reviewed publications, starting with a journal manuscript on the framework planned for 2027.

11. Decisions Requested

Table 57. Decisions requested from the relevant authorities

No.	Decision	Responsible authority	Timing
D1	Accept this Blueprint as a technical input to the national AI policy drafting process	Ministry of Posts, Telecommunications and Information Technology (ICT Division)	Q4 2026
D2	Approve a 12-month Phase 1 pilot, name a government focal point, agree an MoU with a host institution, and commit or identify Year 1 funding (about Tk 4.7 crore, Table 50)	ICT Division and host institution	Within 90 days
D3	Authorise structured bulk access to already-public document collections and nominate data custodians	Legislative and Parliamentary Affairs Division, Finance Division, Planning Commission, BBS, a2i	Within 120 days
D4	Nominate Steering Board members, and agree that ERRC members are selected through an open call by an independent nominations panel	ICT Division and host institution (Steering Board); independent panel (ERRC)	Within 90 days
D5	Commission the feasibility and costing study that sets the Phase 2 budget	ICT Division or a development partner	By month 6
D6	Confirm guardrails: no deployment beyond class A without a G4 decision, and no use in prohibited categories	Steering Board	At charter adoption

Note. None of these decisions creates a statutory body or gives BAPIL legal powers. The approvals and agreements needed before month 1 are listed in Annex T (Table 65).

Resources

Technical Blueprint DOI: DOI to be assigned on publication
White Paper DOI: https://doi.org/10.5281/zenodo.21363967
Executive Summary DOI: https://doi.org/10.5281/zenodo.21441172
GitHub Repository: https://github.com/md-naim-hassan-saykat/bapil

References

Reference numbers [1] to [83] follow the bibliography of the BAPIL White Paper (Version 1.0); new references begin at [84]. Only references cited in this document are listed.

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